Yuval Itzhaki

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https://yuvalyitz.github.io

Appointments

Postdoctoral Researcher | Technische Universität Hamburg 2026.09 - Present
Hamburg, Germany
- Institute for Algorithms and Complexity, working with Prof. Matthias Mnich.

Postdoctoral Researcher | Ben–Gurion University of the Negev 2026.04 - 2026.08
Be'er Sheva, Israel

Education

Department of Industrial Engineering, Ben–Gurion University of the Negev Be'er Sheva, Israel
Ph.D. in Algorithms and Operations Research 2022.10 - 2026.03
- Thesis Title: Scheduling with Intervals
- Advisors: Prof. Danny Hermelin & Prof. Dvir Shabtay
- Research area: Algorithms, Parameterized Complexity, Operations Research.

Department of Computer Science, Technische Universität Berlin Berlin, Germany
M.Sc. in Computer Science 2020.04 - 2021.09
- Thesis Title: Exploiting Order to Find Independent-Sets in Temporal Interval Graphs
- Advisors: Prof. Danny Hermelin, Prof. Rolf Niedermeier
- Majors: Algorithms and Complexity, Data Science

Department of Civil Engineering, Technische Universität Berlin Berlin, Germany
B.Sc. in Civil Engineering 2014.10 - 2018.03
- Majors: Computer Science, Structural Analysis.
- Completed an academic year at RMIT University, Melbourne, Australia.

Honors and Awards

- Kreitman School Dean's List Certificate of Excellence (PhD Studies Award) 2025
- Kreitman Zin doctoral scholarship for outstanding students 2024
- Bertie Lubner prize for Community Service 2025
- Best Poster Award, IE&M Research Day, Ben–Gurion University of the Negev 2026

Publications

1. Heeger, K., Hermelin, D., **Itzhaki, Y.**, Schieber, B., and Shabtay, D. *Just-in-Time Scheduling in Two-Stage Flexible Flow Shops*. *European Journal of Operational Research*, Volume 333, Issue 3, Pages 652–664, 2026. DOI: 10.1016/j.ejor.2026.02.017. Lean 4 formalization.
2. Heeger, K., Hermelin, D., **Itzhaki, Y.**, Molter, H., and Shabtay, D. *Fair Repetitive Interval Scheduling*. *Algorithmica*, Volume 87, Pages 1340–1368, 2025. DOI: 10.1007/s00453-025-01322-y. Lean 4 formalization.
3. Hermelin, D., **Itzhaki, Y.**, Molter, H., and Shabtay, D. *On the Parameterized Complexity of Interval Scheduling with Eligible Machine Sets*. *Journal of Computer and System Sciences*, Volume 144, Article 103533, 2024. DOI: 10.1016/j.jcss.2024.103533. Lean 4 formalization.
4. Hermelin, D., **Itzhaki, Y.**, Molter, H., and Niedermeier, R. *Temporal Interval Cliques and Independent Sets*. *Theoretical Computer Science*, Volume 961, Article 113885, 2023. DOI: 10.1016/j.tcs.2023.113885.

Conference Proceedings

1. Hermelin, D., **Itzhaki, Y.**, Molter, H. and Shabtay, D., Hardness of Interval Scheduling on Unrelated Machines. *Proceedings of the 17th International Symposium on Parameterized and Exact Computation (IPEC)*, 2022.
2. Hermelin, D., **Itzhaki, Y.**, Molter, H. and Niedermeier, R., Temporal Unit Interval Independent Sets. *Proceedings of the 1st Symposium on Algorithmic Foundations of Dynamic Networks (SAND)*, 2022.

Talks

1. Hermelin, D., **Itzhaki, Y.**, Molter, H., and Shabtay, D., Interval Scheduling with Machine Eligibility. *Operations Research Society of Israel (ORSIS) Annual Meeting*, Haifa, 2026.
2. Heeger, K., Hermelin, D., **Itzhaki, Y.**, Schieber, B., and Shabtay, D., Just-in-Time Scheduling in Two-Stage Flexible Flow Shops. *Operations Research Society of Israel (ORSIS) Annual Meeting*, Tel Aviv, 2025.
3. Heeger, K., Hermelin, D., **Itzhaki, Y.**, Molter, H., and Shabtay, D., Fair Repetitive Interval Scheduling. *Operations Research Society of Israel (ORSIS) Annual Meeting*, Be'er Sheva, 2024.

Posters

1. **Itzhaki, Y.** and Mnich, M., Multi-Depot Metric TSP. Poster session at *Mind-IL 2026, Computer Science Conference in Memory of Michael Rabin*, Tel Aviv, 2026.
2. Hermelin, D., **Itzhaki, Y.**, Molter, H., and Shabtay, D., Interval Scheduling with Machine Eligibility. Poster session at *Israel Data Science and Statistics Association Annual Conference*, Tel Aviv, 2026.
3. Hermelin, D., **Itzhaki, Y.**, Molter, H., and Shabtay, D., Interval Scheduling with Eligible Machines. Poster Session at *Pioneering Israeli Science, Science Abroad*, New York, 2026.
4. Hermelin, D., **Itzhaki, Y.**, and Shabtay, D., Scheduling Intervals. **Awarded Best Poster** at the Department of Industrial Engineering and Management Research Day, Ben–Gurion University, 2026.

Teaching

Technische Universität Hamburg — Hamburg, Germany
Institute for Algorithms and Complexity — 2026.09 - Present

- Seminar "Algorithms in the Wild", organizer with Prof. Matthias Mnich, Winter 2026/27.
- "Algorithms and Data Structures", tutorials, Winter 2026/27.

Ben–Gurion University of the Negev — Be'er Sheva, Israel
Department of Industrial Engineering and Management — 2022.10 - 2026.08

- "Simulation", undergraduate course, taught during my postdoc, 2026.
- "Foundations of Algorithms and Complexity", undergraduate course, teaching assistant, 2022–2026.
- Developed structured lecture notes, exercises with solutions, and recorded video lectures for Foundations of Algorithms and Complexity.

Projects

Lean Scheduling Zoo | Code
Scheduling complexity results linked to their Lean formalization. — 2026.09 - Present

Lean 4 formalizations on Lax
Just-in-Time Scheduling in Two-Stage Flexible Flow Shops, *Fair Repetitive Interval Scheduling*, *and Interval Scheduling with Eligible Machine Sets*.

Professional Experience

Data Scientist | Metabuild — 2019.01 - 2023.08
Berlin, Germany

- Developed data pipelines for construction cost estimation from building design models, supporting optimization of sustainable building design.
- Built an end-to-end cost estimation system, from data collection and processing to regression modeling, generating bills of quantities from enriched geometric input.
- Designed and maintained a modular backend architecture in Rust and PostgreSQL, replacing a prototype stack and improving scalability, performance, and maintainability.

Residential Engineer | Intrax Engineering — 2017.09 - 2018.01
Melbourne, Australia

- Designed residential foundations and structures.
- Automated soil data processing via a Netsuite Java module linking reports to design spreadsheets, saving engineers significant time.

Professional Service

Journal reviewer for Discrete Optimization, Journal of Optimization Theory and Applications, and Omega: The International Journal of Management Science.

Conference reviewer for the European Symposium on Algorithms (ESA), the Symposium on Theoretical Aspects of Computer Science (STACS), and the International Symposium on Parameterized and Exact Computation (IPEC).

Community Service

Community Education Center | The Garden Library | Tel Aviv, Israel
Teaching Python, Computer Skills and Hebrew. 2024.02 - 2026.08

Ort Ge'ula High School | Tel Aviv, Israel
Prepared students for Israel's Bagrut exam in Computer Networks. 2024.02 - 2024.06

Skills

Languages: Hebrew (native), English (fluent), German (fluent), French, Arabic.

Programming: `Python`, `SQL`, `Rust`, `Javascript`, `Java`.